

Mitigating Forgetting in Split Learning via Knowledge Distillation

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Abstract—Split learning is a privacy-preserving distributed training paradigm in which a neural network is partitioned between clients and a server, allowing raw data to remain local. Under sequential participation with heterogeneous client data, however, the shared server is updated only by the active client and can drift toward recently observed client distributions, which may reduce accuracy on earlier clients and increase forgetting over the training sequence. We study this setting and propose a server-side distillation method in which the student is regularized by an exponential moving average teacher after a warm-start period. We evaluate the method on CIFAR-10 and CIFAR-100 under sequential heterogeneous client participation. Under strong heterogeneity, the proposed method improves final accuracy by up to 13.7 percentage points on CIFAR-10 and 19.0 percentage points on CIFAR-100, while also improving retention on earlier clients.

I. INTRODUCTION

SPLIT learning (SL) [1] enables collaborative model training without centralizing raw data, making it attractive in settings where privacy, regulation, or institutional boundaries prevent direct data sharing [2]. In SL, a neural network is partitioned into a client-side front model and a server-side back model. Each client maps raw input x to an intermediate activation z and transmits only z to the server. The server completes the forward pass, computes the loss using labels, and returns the cut-layer gradient $\partial L/\partial z$ so the client can update locally by backpropagation. Recent reviews and framework-oriented studies have examined SL from broader system and protocol perspectives, including communication, privacy, deployment trade-offs, and design choices such as label sharing, model aggregation, and cut-layer selection [3], [4].

In practical deployments, clients may participate sequentially rather than in parallel, and their local data may be highly heterogeneous. In that setting, the shared server is updated using gradients from one active client at a time. Once a client finishes training, no further correction arrives for that client distribution. This can shift the shared server toward later clients and reduce performance on earlier ones.

Prior SL work has mainly emphasized privacy, communication efficiency, and protocol design, while retention under strictly sequential non-IID client participation has received less direct attention, especially in settings where a single shared server is trained across heterogeneous clients without replay or

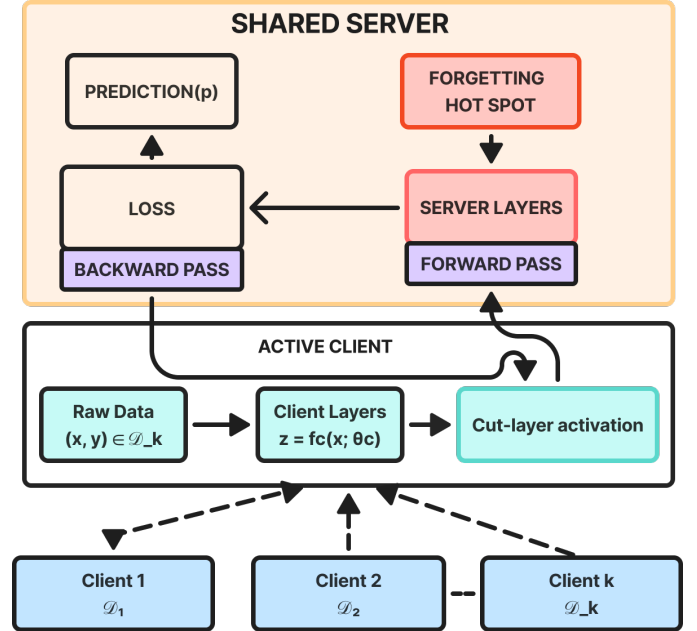


Fig. 1. Split learning overview and sequential forgetting. The active client maps raw input x to a cut-layer activation z and sends z to the shared server. The server computes the loss and returns $\partial L/\partial z$ for client-side backpropagation.

continued access to earlier client data. Recent work has begun to explore more realistic split-learning settings, including rehearsal-based or continual split-learning variants [5], [6], [7]. These directions are relevant, but they do not directly address the setting studied here: a single shared server trained across a fixed sequence of heterogeneous clients without replaying earlier client data. This creates a practical need for a stabilization mechanism that can reduce server drift across client transitions without requiring rehearsal buffers, auxiliary memory, or modifications to the standard client-server exchange.

In this paper, we focus on that failure mode and refer to it as *sequential forgetting in split learning*. Our method operates only on the server side and preserves the standard SL message pattern. The server-side student is trained with supervised loss, while an exponential moving average (EMA) teacher provides softened targets for knowledge distillation (KD) after a warm-start period [8], [9], [10], [11]. The method is designed to reduce abrupt server drift across client transitions without changing the client-server interface.

Contributions.

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- We study split learning with sequential heterogeneous clients and formalize forgetting in this setting using three measurements: accuracy-based forgetting, loss-based forgetting, and representation drift.
- We propose a server-side EMA-guided distillation method with warm-started KD that preserves the standard split-learning interface.
- We evaluate the proposed method on CIFAR-10 and CIFAR-100 under heterogeneous client partitions and show that EMA, KD timing, and learning-rate scheduling all contribute to the final gains.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the problem formulation and proposed approach. Section IV describes the experimental setup, main results, and ablation studies. Section V concludes the paper. The detailed convergence analysis is provided in Appendix A.

II. RELATED WORK

A. Split Learning under Sequential or Evolving Training

Split learning (SL) partitions a neural network between clients and a server so that raw data remain local and only intermediate activations and cut-layer gradients are exchanged during training [1]. Recent reviews organize the growing SL literature around design choices such as cut-layer placement, label sharing, aggregation strategy, and communication-accuracy trade-offs [3]. More recent work has also aimed to unify different SL protocols under a common framework, making comparisons across implementations and deployment settings more systematic [4].

Most early SL studies focused on privacy, efficiency, and protocol design rather than forgetting during sequential training. Hybrid variants such as SplitFed extend SL to multi-client training with federated-style aggregation [12], while other work studies issues such as privacy leakage and changing client availability [6]. These studies are useful for understanding practical SL deployment, but they do not directly study retention of previously learned knowledge under sequential heterogeneous clients.

More recent work has begun to address that gap. PRCSL introduces a rehearsal-based continual split learning framework and reports improved retention under evolving data streams [7]. Split-based Continual Learning (SCL) also combines split learning and continual learning, with catastrophic forgetting treated as a central challenge in model maintenance under changing data [?]. More closely related to our setting, SFedKD studies catastrophic forgetting caused by data heterogeneity in sequential federated training with a server that processes clients in sequence, and mitigates it through multi-teacher knowledge distillation [?].

Our work is closest in spirit to this recent line of research, but differs in both scope and design. We study forgetting in a standard split-learning setting with a shared server trained over a fixed sequence of heterogeneous clients. Rather than relying on rehearsal or multi-teacher selection from past client models, we use a lightweight EMA teacher to stabilize the shared server during client-to-client transitions while keeping the standard split-learning interface unchanged.

B. Knowledge Distillation and EMA-Based Stabilization

Knowledge distillation (KD) trains a student model using both hard labels and softened outputs from a teacher model [10]. Recent surveys show that KD is now used well beyond model compression, including as a regularizer for training stability and representation transfer [11]. In most of this literature, however, the main objective is improved generalization or compactness rather than retention under sequential client updates.

A practical question in KD is how to obtain a teacher that is stable enough to provide useful guidance throughout training. Exponential moving average (EMA) teachers address this by maintaining a slowly updated copy of the student weights. Recent analyses study the dynamics and benefits of EMA in modern deep learning pipelines and provide guidance on how EMA should be configured in practice [8], [9].

Our use of KD differs from the standard setting in two ways. First, the teacher is not introduced for compression, but for stabilizing the shared server-side model in split learning. Second, the goal is not only better final accuracy, but also better retention as training moves across heterogeneous clients. In our method, the EMA teacher serves as a stable reference for the server, and distillation is introduced only after a warm-start period so that early supervised learning is not over-constrained.

III. PROBLEM FORMULATION AND APPROACH

A. Sequential client participation in split learning

We consider split learning with K clients participating sequentially. Client k holds a local dataset $\mathcal{D}_k$ with size $n_k = |\mathcal{D}_k|$, and the total number of samples is

$$N = \sum_{k=1}^K n_k. \quad (1)$$

The model is partitioned into a client-side segment $f_c(\cdot; \theta_c)$ and a server-side segment $f_s(\cdot; \theta_s)$, with parameters θ_c and θ_s , respectively.

For an input-label pair (x, y) , the client computes the cut-layer activation

$$z = f_c(x; \theta_c), \quad (2)$$

and sends z to the server. The server then computes logits

$$s = f_s(z; \theta_s), \quad (3)$$

followed by predicted probabilities

$$p = \text{softmax}(s). \quad (4)$$

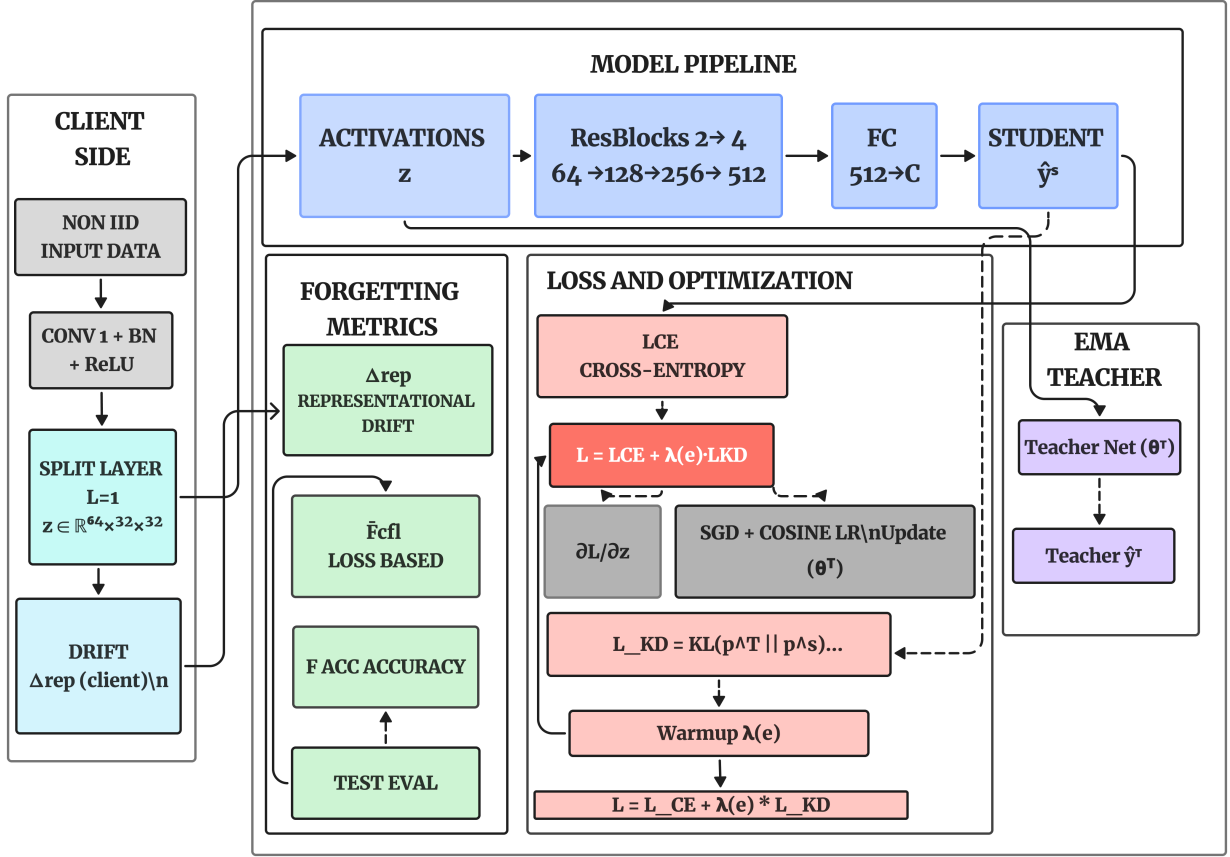


Fig. 2. Overview of the proposed training pipeline. The client computes cut-layer activations z and sends them to the server. The server-side student is trained using cross-entropy and, after a warm-start period, a distillation loss based on soft targets from an EMA teacher.

The supervised objective is the cross-entropy loss

$$L_{CE}(p, y) = - \sum_{c=1}^C \mathbb{1}[y = c] \log p_c, \quad (5)$$

where C is the number of classes and $\mathbb{1}[\cdot]$ is the indicator function.

Training is organized as a sequence of client stages, with only one client active at any given time and the shared server updated exclusively from that client's data during its stage. During the stage corresponding to client k , the shared server parameters are updated exclusively using data from $\mathcal{D}_k$. Once client k is no longer active, the server receives no further gradient signal from that client distribution. Under heterogeneous client data, later updates can therefore reduce performance on earlier clients. In standard split learning, the server updates θ_s using the supervised loss and returns the cut-layer gradient $\partial L / \partial z$ to the client, which then updates θ_c . Under sequential participation, the shared server-side model is repeatedly optimized using supervision from only the currently active client. As a result, the server representation can gradually shift toward recently observed clients, which motivates a

stabilization mechanism for the shared server-side model that does not change the standard split-learning interface.

B. Server-side stabilization with EMA-guided distillation

We address this problem by regularizing the shared server during sequential training. The server-side student is trained with supervised loss, while an exponential moving average teacher provides softened targets for distillation after a warm-start period. This leaves the standard SL message pattern unchanged. Because the server is updated by one client distribution at a time, its parameters can move sharply toward the active client. The EMA teacher averages recent server states and therefore provides a smoother target than the current student alone [8], [9]. Distillation toward this target penalizes abrupt changes in server outputs on current activations and helps reduce short-horizon drift across client transitions.

We use knowledge distillation so that the server-side student is trained not only with the hard labels of the current client but also with the teacher's softened predictions [10], [11]. Fig. 2 shows the overall training pipeline, including the client-side

split point, the server-side student, the EMA teacher, and the combined supervised and distillation losses.

Let s denote the student logits and t denote the teacher logits for the same activation z . Let $T > 0$ be the temperature parameter used to soften the output distributions. The softened student and teacher distributions are defined as

$$p^{(T)} = \text{softmax}(s/T), \quad q^{(T)} = \text{softmax}(t/T). \quad (6)$$

The distillation loss is

$$L_{KD} = T^2 \text{KL}(q^{(T)} \| p^{(T)}), \quad (7)$$

where $\text{KL}(\cdot \| \cdot)$ denotes the Kullback–Leibler divergence. KD is applied only on the current client’s activations z ; the method does not replay earlier client activations or raw data.

The EMA teacher is updated from the student server parameters. Let θ_t^e denote the teacher parameters at epoch e , and let θ_s^e denote the student server parameters at epoch e . The EMA update is given by

$$\theta_t^e = \beta \theta_t^{e-1} + (1 - \beta) \theta_s^e, \quad (8)$$

where $\beta \in (0, 1)$ is the EMA decay coefficient. In our implementation, we use $\beta = 0.99$ so that the teacher evolves more slowly than the student and acts as a stable reference during training.

A practical issue is when distillation should begin and how strongly it should influence training. If it is applied too early, it may constrain the student before a useful supervised representation has formed. To address this, we introduce a warm-start period of w epochs during which distillation is disabled. After warm-start, the total loss is defined as

$$L = (1 - \lambda_e) L_{CE} + \lambda_e L_{KD}, \quad (9)$$

where $\lambda_e \in [0, \lambda_{\max}]$ is an epoch-dependent distillation weight. We use a progressive schedule in which distillation is zero during warm-start and then increases gradually:

$$\lambda_e = \begin{cases} 0, & e \leq w, \\ \lambda_{\max} \min\left(1, \frac{e - w}{E_{\text{train}} - w}\right), & e > w. \end{cases} \quad (10)$$

Here, e denotes the local epoch within the current client stage. When training shifts to a different client, both the warm-start phase and the distillation schedule are reset. This design keeps early training label-driven and introduces stronger distillation only after the student and teacher have become informative.

C. Training procedure

Algorithm 1 summarizes the full training procedure. We train across K clients in a fixed sequential order. Client parameters are local to each client, while the server parameters are shared across the client sequence. For each client, training proceeds over E_{train} epochs using mini-batches of size B . In each mini-batch, the client computes the cut-layer activation z and sends it to the server. The server computes the supervised loss and, when distillation is active, the KD loss using the EMA teacher. The resulting total loss is used to update

Algorithm 1 Sequential Split Learning with Warm-Started EMA Distillation

Require: Sequential clients $\{1, \dots, K\}$ with datasets $\{\mathcal{D}_k\}_{k=1}^K$; epochs per client E_{train} ; mini-batch size B . Client model $f_c(\cdot; \theta_c)$; server student model $f_s(\cdot; \theta_s)$; teacher parameters $\theta_t \leftarrow \theta_s$. Temperature T ; EMA decay β ; warm-start w ; maximum distillation weight $\lambda_{\max}$; learning rate η .

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1: for  $k = 1$  to  $K$  do
2:   for  $e = 1$  to  $E_{\text{train}}$  do
3:     if  $e \leq w$  then
4:        $\lambda_e \leftarrow 0$ 
5:     else
6:        $\lambda_e \leftarrow \lambda_{\max} \min\left(1, \frac{e - w}{E_{\text{train}} - w}\right)$ 
7:     end if
8:     for each mini-batch  $(x, y) \in \mathcal{D}_k$  do
9:       Client forward:  $z \leftarrow f_c(x; \theta_c)$ ; send  $z$  to server
10:      Student forward:  $s \leftarrow f_s(z; \theta_s)$ 
11:       $L_{CE} \leftarrow \text{CE}(\text{softmax}(s), y)$ 
12:      if  $\lambda_e > 0$  then
13:        Teacher forward:  $t \leftarrow f_s(z; \theta_t)$ 
14:         $p \leftarrow \text{softmax}(s/T)$ ;  $q \leftarrow \text{softmax}(t/T)$ 
15:         $L_{KD} \leftarrow T^2 \text{KL}(q \| p)$ 
16:         $L \leftarrow (1 - \lambda_e) L_{CE} + \lambda_e L_{KD}$ 
17:      else
18:         $L \leftarrow L_{CE}$ 
19:      end if
20:      Server update:  $\theta_s \leftarrow \theta_s - \eta \nabla_{\theta_s} L$ 
21:      Send  $\partial L / \partial z$  to client; update  $\theta_c$  by backpropagation
22:    end for
23:    EMA update:  $\theta_t \leftarrow \beta \theta_t + (1 - \beta) \theta_s$ 
24:  end for
25: end for

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the server-side student model. The corresponding cut-layer gradient is then returned to the client so that the client-side model can be updated by backpropagation. At the end of each epoch, the teacher parameters are updated using the EMA rule in Eq. (8).

D. Retention measurements

To evaluate forgetting beyond final accuracy, we track three complementary signals over the client sequence. Let E denote the final checkpoint index.

Let $A_k^{(e)}$ denote test accuracy on client k after checkpoint e . Accuracy-based forgetting is

$$F_k^{\text{acc}} = \max_e A_k^{(e)} - A_k^{(E)}. \quad (11)$$

Let $h^{(e)}(x)$ denote the cut-layer activation produced for reference sample x at checkpoint e . Loss-based forgetting is

$$F^{\text{loss}} = \frac{1}{K} \sum_{k=1}^K \left(L(\theta^{(E)}; T_k) - \min_e L(\theta^{(e)}; T_k) \right). \quad (12)$$

Let $h^{(e)}(x)$ denote the selected representation for reference sample x at checkpoint e . Representation drift is

$$\Delta_{\text{rep}}^{(e)} = \frac{1}{|\mathcal{X}_{\text{ref}}|} \sum_{x \in \mathcal{X}_{\text{ref}}} \|h^{(e)}(x) - h^{(e-1)}(x)\|_2. \quad (13)$$

We also study the optimization behavior of the proposed EMA-guided server-side update under standard smoothness,

TABLE I
FINAL TEST ACCURACY AND LOSS. BEST RESULTS ARE BOLDED.

Setting	Method	Acc. (%)	Loss
C10, $\alpha = 0.5$	Vanilla SL	76.81	0.8245
	Static-teacher KD	69.34	1.1256
	EMA-KD	72.18	0.9837
	Proposed	79.80	0.6986
C10, $\alpha = 0.1$	Vanilla SL	40.18	2.1467
	Static-teacher KD	28.45	2.8834
	EMA-KD	31.92	2.6521
	Proposed	53.88	1.8533
C100, $\alpha = 0.5$	Vanilla SL	50.62	2.1695
	EMA-KD	29.76	3.4958
	Proposed	61.83	1.7906
C100, $\alpha = 0.1$	Vanilla SL	27.90	3.4249
	EMA-KD	14.99	4.4143
	Proposed	46.90	2.7084

bounded-variance, and bounded-heterogeneity assumptions. The resulting convergence guarantee and its derivation are provided in Appendix A.

IV. EXPERIMENTAL SETUP AND MAIN RESULTS

A. Experimental setup

We evaluate the method on CIFAR-10 and CIFAR-100 using a split ResNet-18 model, with the split placed after the first layer. The data is partitioned across K clients that train sequentially in a fixed order. To control client heterogeneity, we use Dirichlet partitioning with $\alpha \in \{0.5, 0.1\}$, where smaller α gives a more uneven class distribution across clients.

For CIFAR-10, we train for 100 epochs with batch size 128. For CIFAR-100, we train for 200 epochs with batch size 64. The proposed method follows the EMA-guided distillation procedure described in Section III, including warm-started knowledge distillation and the final learning-rate schedule used in our implementation. We keep the architecture, client order, and data partitions fixed across all methods.

We compare four methods: *Vanilla SL*, which uses only the supervised cross-entropy loss; *Static-teacher KD*, which applies knowledge distillation with a fixed teacher; *EMA-KD*, which uses an EMA teacher without the full warm-started progressive design; and *Proposed*, which combines EMA guidance, warm-started knowledge distillation, and the final optimization schedule.

B. Final performance

Table I reports final test accuracy and loss for all settings. The proposed method gives the highest final accuracy and the lowest final loss in every reported case. The gain is limited under $\alpha = 0.5$ and much larger under $\alpha = 0.1$.

On CIFAR-10 with $\alpha = 0.5$, final accuracy increases from 76.81% under Vanilla SL to 79.80%. For $\alpha = 0.1$, it increases from 40.18% to 53.88%. On CIFAR-100, the same pattern appears: for $\alpha = 0.5$, the proposed method reaches 61.83%, compared with 50.62% for Vanilla SL, and for $\alpha = 0.1$, it reaches 46.90%, compared with 27.90%.

TABLE II
COMPACT FORGETTING SUMMARY. LOWER IS BETTER.

Setting	Method	Avg. F^{acc}	Avg. Δ_{rep}
C10, $\alpha = 0.5$	Vanilla SL	5.8468	88.6493
	Static-teacher KD	1.2370	–
	EMA-KD	2.4656	41.8072
	Proposed	1.1763	54.2126
C10, $\alpha = 0.1$	Vanilla SL	4.9607	62.1225
	Static-teacher KD	1.2370	–
	EMA-KD	5.1578	54.4275
	Proposed	2.0092	59.0056
C100, $\alpha = 0.5$	Vanilla SL	1.3251	60.2491
	Static-teacher KD	1.2987	–
	EMA-KD	1.1110	36.9855
	Proposed	5.4510	35.5656
C100, $\alpha = 0.1$	Vanilla SL	2.7268	68.0320
	Static-teacher KD	1.2987	–
	EMA-KD	0.5502	36.3284
	Proposed	9.1078	35.9750

Fig. 3 shows the test-accuracy trajectories for all four settings. Consistent with Table I, the proposed method reaches the highest final accuracy in every case, and the gap is largest under $\alpha = 0.1$. In those two settings, Vanilla SL improves more slowly and plateaus earlier.

C. Retention and forgetting

Final accuracy does not by itself show how much earlier clients are forgotten during sequential training. To examine that part of the behavior, Table II reports two quantities: accuracy-based forgetting and representation drift. The retention results are less uniform than the final-accuracy results. On CIFAR-10, the proposed method lowers accuracy-based forgetting relative to Vanilla SL in both heterogeneity settings. On CIFAR-100, the picture is mixed: EMA-KD gives the lowest accuracy-based forgetting, while the proposed method gives the lowest representation drift. This means the proposed method is strongest on final predictive performance, but not uniformly best on every forgetting-related quantity.

Overall, the proposed method is the strongest on final accuracy across all four settings. Its effect on forgetting is more dependent on the dataset and the metric being measured. On CIFAR-10, the gains extend to both final performance and retention. On CIFAR-100, the method still gives the best final model, but the forgetting metrics are less consistent.

D. Ablation studies

We next look at which parts of the method contribute to the final result. The ablation comparison focuses on three changes: adding teacher guidance, replacing a fixed teacher with an EMA teacher, and introducing warm-started scheduling.

Table III reports the component-wise comparison on CIFAR-100 with $\alpha = 0.1$, which is the hardest setting in our experiments. Vanilla SL uses only supervised training. Static-teacher KD adds teacher guidance without EMA smoothing. EMA-KD adds the EMA teacher but not the full warm-started design. The proposed method combines all components.

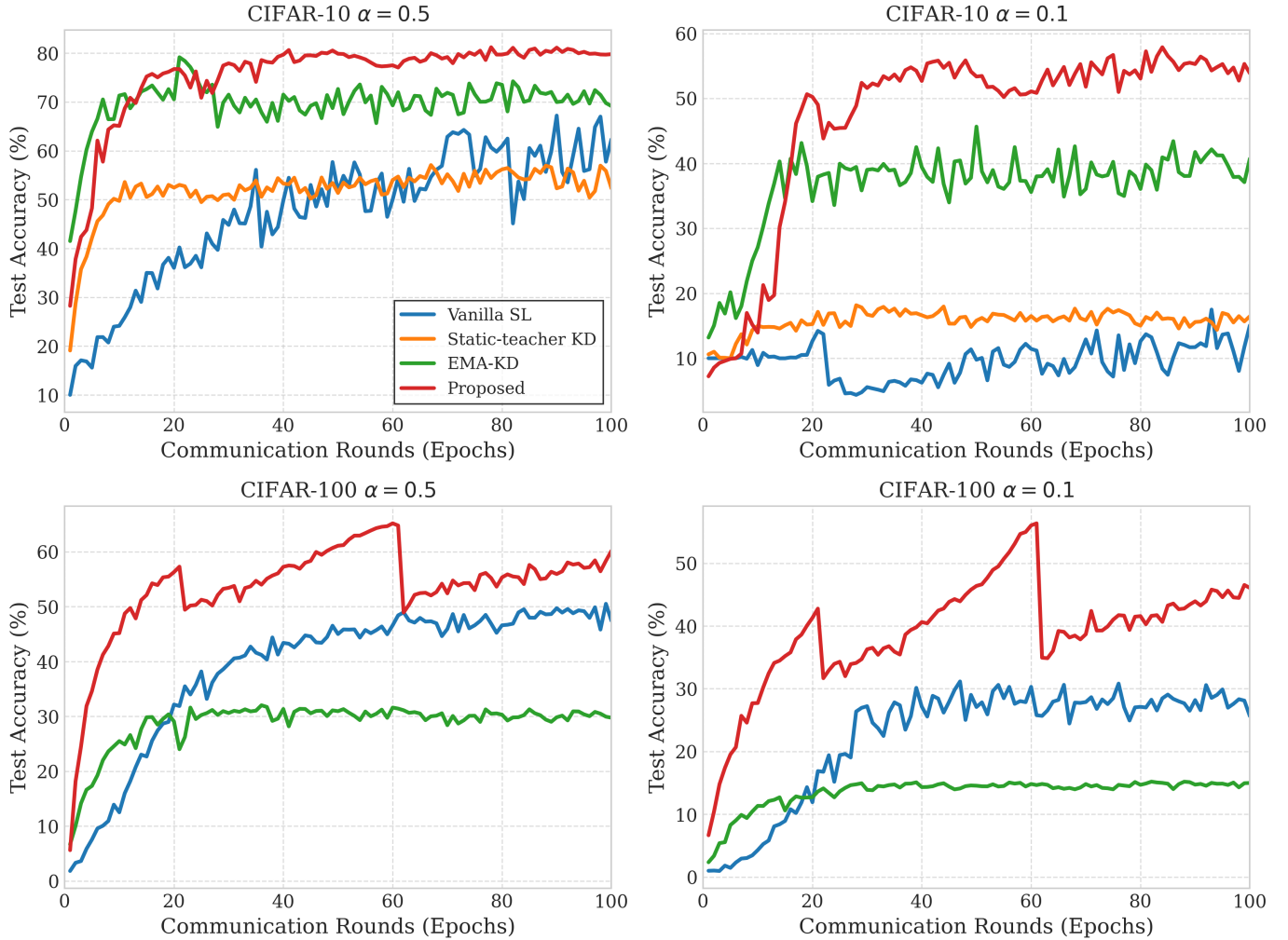


Fig. 3. Test-accuracy trajectories across datasets and heterogeneity levels. The proposed method reaches the best final accuracy in all four settings, with the clearest gap under stronger heterogeneity.

TABLE III
COMPONENT ABLATION ON CIFAR-100 WITH $\alpha = 0.1$. LOWER IS BETTER FOR FORGETTING AND REPRESENTATION DRIFT.

Variant	Final Acc.	Avg. F^{acc}	Avg. Δ_{rep}
Vanilla SL	27.90	2.7268	68.0320
Static-teacher KD	—	1.2987	—
EMA-KD	14.99	0.5502	36.3284
Proposed	46.90	9.1078	35.9750

The ablation results do not point to a single component being responsible for the full gain. EMA-KD gives the lowest accuracy-based forgetting in this setting, which suggests that the EMA teacher directly helps retention. The full method, however, gives the highest final accuracy and the lowest representation drift. This indicates that warm-starting and the final optimization schedule matter for the quality of the final server model, even if they do not minimize every forgetting metric.

Fig. 4 shows the same comparison graphically. The ablation results indicate that no single component explains the full

gain. Teacher guidance, EMA smoothing, and warm-started scheduling each affect training, and the full combination gives the strongest final result in the hardest setting.

V. CONCLUSION

We studied split learning with heterogeneous clients participating in sequence, where the shared server is updated by one client at a time and performance on earlier clients can deteriorate as training proceeds. To address this problem, we introduced a server-side distillation method that combines an EMA teacher with warm-started knowledge distillation, while keeping the standard split-learning protocol unchanged.

Experiments on CIFAR-10 and CIFAR-100 showed that the proposed method consistently improves final accuracy, with the largest gains appearing under stronger heterogeneity. The results also showed that its effect on forgetting is more setting-dependent: on CIFAR-10, the method improves both final performance and retention, while on CIFAR-100 it gives the strongest final model even though forgetting metrics are less uniform. The ablation study further showed that the full gain

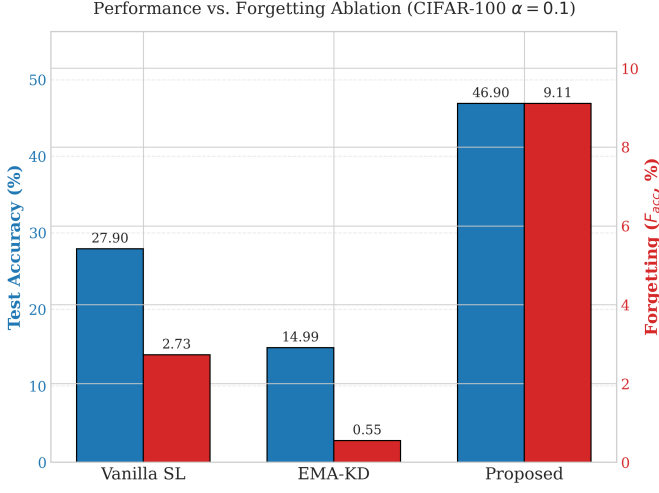


Fig. 4. Ablation comparison on CIFAR-100 with $\alpha = 0.1$. The full method gives the best final accuracy, while EMA-based variants reduce representation drift relative to Vanilla SL.

does not come from a single component alone, but from the combination of teacher guidance, EMA smoothing, and warm-started scheduling.

Overall, these results show that stabilizing the shared server is an effective way to improve sequential split learning under heterogeneous client participation, without replaying earlier client data or changing the client-server interface.

APPENDIX

This appendix provides the full derivation used to analyze the proposed EMA-guided server-side distillation update. It serves as the detailed mathematical companion to the summary statement given in the main paper.

For consistency with the proof draft, we retain its notation and intermediate steps in the presentation below.

A. Full Proof

Our algorithm can be integrated into the previous proof.

Using the L -smoothness assumption, we can state that

$$\mathbb{E}[f(\theta + \Delta x) - f(\theta)] \leq \mathbb{E}[\langle \nabla f, \Delta x \rangle] + \frac{L}{2} \mathbb{E}[\|\Delta x\|^2] \quad (14)$$

Then we have K as the total number of participating clients, η = learning rate, L = the number of total local update steps, and m = index of clients, so

$$\mathbb{E}\langle \nabla f, \Delta x \rangle = -\eta KL \left(\mathbb{E} \left[\langle \nabla f, \frac{1}{K} \sum_{m=1}^K \frac{1}{L} \sum_{l=1}^L (\nabla f_{l_m})(x_{m,l}) - \nabla f_{l_m}(x) + \nabla f_{l_m}(x) \rangle \right] \right) \quad (15)$$

Note that the second part of (2) is generated from substituting the expression for Δx in. Then, we simplify that expression by taking out

$$\mathbb{E}[\nabla f_{l_m} * \nabla f(x)] = \|\nabla f(x)\|^2 \quad (16)$$

to get

$$\begin{aligned} & -\eta KL \|\nabla f(x)\|^2 - \eta KL \left(\mathbb{E} \left[\langle \nabla f(x), \frac{1}{K} \sum_{m=1}^K \frac{1}{L} \sum_{l=1}^L (\nabla f_{l_m})(x_{m,l}) - \nabla f_{l_m}(x) \rangle \right] \right) \\ & \leq -\eta KL \|\nabla f(x)\|^2 + \frac{1}{2} \eta KL \|\nabla f(x)\|^2 + \frac{1}{2} \eta \left(\sum_{m=1}^K \sum_{l=1}^L \mathbb{E}[(\nabla f_{l_m})(x_{m,l}) - \nabla f_{l_m}(x)]^2 \right) \\ & = -\frac{1}{2} \eta KL \|\nabla f(x)\|^2 + \frac{1}{2} (L_f)^2 \eta \left(\sum_{m=1}^K \sum_{l=1}^L \mathbb{E}[\|x_{m,l} - x\|^2] \right) \end{aligned} \quad (17)$$

We are able to derive the previous inequality from the fact that

$$|\langle a, b \rangle| \leq \frac{1}{2} a^2 + \frac{1}{2} b^2,$$

or

$$\begin{aligned} 2(a * b) & \leq a^2 + b^2; \\ (a - b)^2 & \geq 0 \end{aligned}$$

From here, we take another component of the smoothness assumption to bound:

$$\begin{aligned} \frac{1}{2} L \mathbb{E} \|\Delta x\|^2 & \leq 2L\eta^2 \left(\mathbb{E} \left\| \sum_{m=1}^K \sum_{l=0}^{L-1} (g_{\pi_m, l} - \nabla f_{\pi_m}(x_{m,l})) \right\|^2 \right) \\ & + 2L\eta^2 \left(\mathbb{E} \left\| \sum_{m=1}^K \sum_{l=0}^{L-1} (\nabla f_{\pi_m}(x_{m,l}) - \nabla f_{\pi_m}(x)) \right\|^2 \right) \\ & + 2L\eta^2 \left(\mathbb{E} \left\| \sum_{m=1}^K \sum_{l=0}^{L-1} \nabla f_{\pi_m}(x) - \nabla f(x) \right\|^2 \right) \\ & + 2L\eta^2 \left(\mathbb{E} \left\| \sum_{m=1}^K \sum_{l=0}^{L-1} (\nabla f(x)) \right\|^2 \right) \end{aligned} \quad (18)$$

In this inequality, we are splitting the update Δx into 4 pieces: the overall gradient $\nabla f(x)$, the difference between any given client's gradient and the overall gradient $\nabla f_{\pi_m}(x) - \nabla f(x)$, the difference between any local update and the overall client gradient $\nabla f_{\pi_m}(x_{m,l}) - \nabla f_{\pi_m}(x)$, and the difference between the stochastic gradient of f_{π_m} and a local update $g_{\pi_m, l} -$

$\nabla f_{\pi_m}(x_{m,l})$.

Now, we will bound each of the 4 parts.

(1) Using the assumption that

$$\mathbb{E} \|\nabla f_m(x; \xi) - \nabla f_m(x)\|^2 \leq \sigma^2 \quad (19)$$

the linearity of expectation which allows us to take out the summations from the LHS, and the calculate the bound multiplied over all K machines and L local update steps, we get the following inequality for the first term:

$$2L\eta^2 \left(\sum_{m=1}^K \sum_{l=0}^{L-1} \mathbb{E} \|g_{\pi_m, l} - \nabla f_{\pi_m}(x_{m,l})\|^2 \right) \leq 2L\eta^2 KL\sigma^2 \quad (20)$$

(2) Next, using the L -smoothness assumption

$$\mathbb{E} \|f_m(x) - f_m(y)\| \leq L\|x - y\| \quad (21)$$

we square both side and get

$$\mathbb{E} \|f_m(x) - f_m(y)\|^2 \leq L^2\|x - y\|^2 \quad (22)$$

Then, given inequality (2) and the linearity of expectation, the following inequality holds for the second term:

$$2L\eta^2 \left(\mathbb{E} \left\| \sum_{m=1}^K \sum_{l=0}^{L-1} \nabla f_{\pi_m}(x_{m,l}) - \nabla f_{\pi_m}(x) \right\|^2 \right) \leq 2L^3\eta^2 \sum_{m=1}^K \sum_{l=0}^{L-1} \mathbb{E} \|x_{m,l} - x\|^2 \quad (23)$$

Multiplying both sides by KL , we get

$$2L\eta^2 KL \left(\mathbb{E} \left\| \sum_{m=1}^K \sum_{l=0}^{L-1} \nabla f_{\pi_m}(x_{m,l}) - \nabla f_{\pi_m}(x) \right\|^2 \right) \leq 2L^3\eta^2 KL \sum_{m=1}^K \sum_{l=0}^{L-1} \mathbb{E} \|x_{m,l} - x\|^2 \quad (24)$$

(3) We see that, in the lemma for sampling without replacement:

$$\mathbb{E} \|x - \bar{x}\|^2 = \frac{n-s}{s(n-1)} \zeta^2 \quad (25)$$

The following inequality holds for the third term:

$$\begin{aligned} & 2L\eta^2 \left(\mathbb{E} \left\| \sum_{m=1}^K \sum_{l=0}^{L-1} \nabla f_{\pi_m}(x) - \nabla f(x) \right\|^2 \right) \\ & \leq 2L\eta^2 KL \sum_{m=1}^K \sum_{l=0}^{L-1} \frac{M-S}{S(M-1)} \\ & = 2L\eta^2 K^2 L^2 \frac{M-S}{S(M-1)} \zeta^2. \end{aligned} \quad (26)$$

Using that same lemma, we can get that

$$2L\eta^2 \left(\mathbb{E} \left\| \sum_{m=1}^K \sum_{l=0}^{L-1} \nabla f(x) \right\|^2 \right) \leq 2L\eta^2 K^2 L^2 \left(1 + \frac{M-S}{S(M-1)} \right) \|\nabla f(x)\|^2 \quad (27)$$

Plugging (7), (11), (13), and (14) back into the original bound, we get that

$$\begin{aligned} \mathbb{E}[f(\theta + \Delta x) - f(\theta)] & \leq -\frac{1}{6}KL\eta \|\nabla f(x)\|^2 \\ & \quad + 2L\eta^2 KL\sigma^2 \\ & \quad + 2L\eta^2 K^2 L^2 \frac{M-S}{S(M-1)} \zeta^2 \\ & \quad + \frac{5}{6}L^2\eta \left(\sum_{m=1}^K \sum_{l=0}^{L-1} \mathbb{E} \|x_{m,l} - x\|^2 \right) \end{aligned} \quad (28)$$

Rearranging the bound for

$$\frac{1}{2}L\mathbb{E} \|\Delta x\|^2, \quad (29)$$

from (5), we get

$$\begin{aligned}
& \mathbb{E} \|\Delta x\|^2 = \mathbb{E} \|x_{m,l} - x\|^2 \\
& \leq 4\eta^2 \left(\mathbb{E} \left\| \sum_{i=1}^m \sum_{j=0}^{b_{m,k}(i)} (g_{\pi_{i,j}} - \nabla f_{\pi_i}(x_{i,j})) \right\|^2 \right) \\
& + 4\eta^2 \left(\mathbb{E} \left\| \sum_{i=1}^m \sum_{j=0}^{b_{m,k}(i)} (\nabla f_{\pi_i}(x_{i,j}) - \nabla f_{\pi_i}(x)) \right\|^2 \right) \\
& + 4\eta^2 \left(\mathbb{E} \left\| \sum_{i=1}^m \sum_{j=0}^{b_{m,k}(i)} (\nabla f_{\pi_i}(x) - f(x)) \right\|^2 \right) \\
& + 4\eta^2 \left(\mathbb{E} \left\| \sum_{i=1}^m \sum_{j=0}^{b_{m,k}(i)} (\nabla f(x)) \right\|^2 \right). \tag{30}
\end{aligned}$$

As Δx is the local update, we can substitute this as $x_{m,l} - x$. Using the linearity of expectation and the same process shown in (7), we see that

$$\begin{aligned}
& 4\eta^2 \left(\mathbb{E} \left\| \sum_{i=1}^m \sum_{j=0}^{b_{m,k}(i)} (g_{\pi_{i,j}} - \nabla f_{\pi_i}(x_{i,j})) \right\|^2 \right) \\
& = 4\eta^2 \sum_{i=1}^m \sum_{j=0}^{b_{m,k}(i)} \left(\mathbb{E} \|(g_{\pi_{i,j}} - \nabla f_{\pi_i}(x_{i,j}))\|^2 \right) \\
& = 4\eta^2 B_{m,k} \mathbb{E} \|(g_{\pi_{i,j}} - \nabla f_{\pi_i}(x_{i,j}))\|^2 \leq 4\eta^2 B_{m,k} \sigma^2 \tag{31}
\end{aligned}$$

where $B_{m,k} = m * (b_{m,k}(i))$.

Next, similar to (10), we see that

$$\begin{aligned}
& 4\eta^2 \left(\mathbb{E} \left\| \sum_{i=1}^m \sum_{j=0}^{b_{m,k}(i)} \nabla f_{\pi_i}(x_{i,j} - \nabla f_{\pi_i}(x)) \right\|^2 \right) \\
& \leq 4\eta^2 L^2 \left(\sum_{i=1}^m \sum_{j=0}^{b_{m,k}(i)} \mathbb{E} \|x_{i,j} - x\|^2 \right) \tag{32}
\end{aligned}$$

For

$$4\eta^2 \left(\mathbb{E} \left\| \sum_{i=1}^m \sum_{j=0}^{b_{m,k}(i)} \nabla(f(x)) \right\|^2 \right) \tag{33}$$

we can use the linearity of expectation as well as the fact that

$$\mathbb{E} \|\nabla f(x)\|^2 = |\nabla f(x)|^2 \tag{34}$$

to simplify (20) to

$$4\eta^2 \left(\sum_{i=1}^m \sum_{j=0}^{b_{m,k}(i)} \|\nabla f(x)\|^2 \right). \tag{35}$$

We will leave the third term of the RHS of the inequality unsimplified for now. So, putting (18), (19), (22), and the third term together, we get

$$\begin{aligned}
& \mathbb{E} \|x_{m,l} - x\|^2 \leq 4\eta^2 B_{m,k} \sigma^2 \\
& + 4\eta^2 L^2 \left(\sum_{i=1}^m \sum_{j=0}^{b_{m,k}(i)} \mathbb{E} \|x_{i,j} - x\|^2 \right)
\end{aligned}$$

$$\begin{aligned}
& +4\eta^2\mathbb{E}||\nabla f_{\pi_i} - f(x)||^2 \\
& +4\eta^2\left(\sum_{i=1}^m\sum_{j=0}^{b_{m,k}(i)}||f(x)||^2\right) + 4\eta^2\mathbb{E}||\sum_{i=1}^m\sum_{j=0}^{b_{m,k}(i)}\nabla f_{\pi_i}(x) - f(x)||^2.
\end{aligned} \tag{36}$$

Bounding (23) using the facts that

$$\sum_{m=1}^K\sum_{l=0}^{L-1}B_{m,k}\leq\frac{1}{2}K^2L^2 \tag{37}$$

and

$$\sum_{m=1}^K\sum_{l=0}^{L-1}B_{m,k}^2\leq\frac{1}{3}K^3L^3 \tag{38}$$

we get

$$\begin{aligned}
E_r & \leq 2K^2L^2\eta^2\sigma^2 + 2L^2K^2L^2\eta^2E_r \\
& + K^2L^3\eta^2\zeta^2 + 2\left(\frac{\beta^2}{K} + 1\right)K^3L^3\eta^2||\nabla f(x)||^2
\end{aligned} \tag{39}$$

Using

$$\eta \leq \frac{1}{6LKL(1+\frac{\beta^2}{K})} \tag{40}$$

we get

$$E_r \leq \frac{9}{4}K^2L^2\eta^2\sigma^2 + \frac{9}{4}K^2L^3\eta^2\zeta^2 + \frac{9}{4}\left(\frac{\beta^2}{K} + 1\right)K^3L^3\eta^2||\nabla f(x)||^2 \tag{41}$$

Now, using the results from (28), we want to re-write (15).

Multiplying the $\frac{5}{6}L^2\eta(\sum_{m=1}^K\sum_{l=0}^{L-1}\mathbb{E}||x_{m,k}-x||^2)$ section of (15) and the $\frac{9}{4}K^2L^2\eta^2\sigma^2$ section of (28) to calculate a more complicated part of the overall bound, we get

$$\frac{5}{6}L^2\eta * \frac{9}{4}K^2L^2\eta^2\sigma^2 = \frac{15}{8}L^2\eta^3K^2L^2\sigma^2. \tag{42}$$

We do this similarly by multiplying the $\frac{5}{6}L^2\eta(\sum_{m=1}^K\sum_{l=0}^{L-1}\mathbb{E}||x_{m,k}-x||^2)$ term from (15) and the $\frac{9}{4}K^2L^3\eta^2\zeta^2$ term from (28) to get

$$\frac{5}{6}L^2\eta * \frac{9}{4}K^2L^3\eta^2\zeta^2 = \frac{15}{8}L^2\eta^3L^2K^3\zeta^2. \tag{43}$$

From the (15) term $-\frac{1}{6}KL\eta||\nabla f(x)||^2$, we get

$$-\frac{1}{6}KL\eta||\nabla f(x)||^2 \leq -\frac{1}{10}KL\eta\mathbb{E}||\nabla f(x^r)||^2. \tag{44}$$

Adding (29), (30), (31), the $2L\eta^2KL\sigma^2$ term and the $2L\eta^2K^2L^2\frac{M-K}{K(M-1)}\zeta^2$ term from (15), we get

$$\begin{aligned}
\mathbb{E}[f(\theta + \Delta x) - f(\theta)] & \leq 2L\eta^2KL\sigma^2 + \frac{15}{8}L^2\eta^3K^2L^2\sigma^2 \\
& + \frac{15}{8}L^2\eta^3L^2K^3\zeta^2 - \frac{1}{10}KL\eta\mathbb{E}||\nabla f(x^r)||^2 \\
& + 2L\eta^2K^2L^2\frac{M-K}{K(M-1)}\zeta^2
\end{aligned} \tag{45}$$

Using (32), and we define

$$\bar{\eta} = KL\eta \tag{46}$$

to get

$$\begin{aligned}
\mathbb{E}(f(x^{r+1}) - f^*) & \leq \mathbb{E}(f(x^r) - f^*) - \frac{\bar{\eta}}{10}\mathbb{E}||\nabla f(x^r)||^2 \\
& + \frac{2L\bar{\eta}^2\sigma^2}{KL} + 2L\bar{\eta}^2\frac{M-K}{K(M-1)}\zeta^2 \\
& + \frac{15}{8}\frac{L^2\bar{\eta}^3\sigma^2}{KL} + \frac{15}{8}\frac{L^2\bar{\eta}^3\zeta^2}{K}
\end{aligned} \tag{47}$$

where we subtracted f^* , or the optimal value of f .

From here we use the fact that if we have 2 non-negative series r_t and s_t where

$$r_{t+1} \leq r_t - b\gamma s_t + c_1\gamma^2 c_2\gamma^3 \quad (48)$$

for $t \geq 0, b > 0, c_1, c_2 \geq 0$ and learning rate γ where

$$0 \leq \gamma \leq \frac{1}{d}, \quad (49)$$

we can set equal weight $w_t = 1$ for all t and $W_T = \sum_{t=0}^T w_t = T$. Then, we define

$$\omega_T = \frac{1}{W_T} \sum_{t=0}^T s_t w_t \leq \frac{r_0}{b\gamma(T+1)} + \frac{c_1\gamma}{b} + \frac{c_2\gamma^2}{b} \quad (50)$$

and we take

$$\gamma = \min \left(\left(\frac{r_0}{c_1 T} \right)^{\frac{1}{2}}, \left(\frac{r_0}{c_2 T} \right)^{\frac{1}{3}}, \left(\frac{1}{d} \right) \right) \quad (51)$$

and we can use (37) and (38) to get

$$\omega_T = \mathcal{O} \left(\frac{dr_0}{T} + \frac{c_1^{\frac{1}{2}} r_0^{\frac{1}{2}}}{T^{\frac{1}{2}}} + \frac{c_2^{\frac{1}{3}} r_0^{\frac{2}{3}}}{T^{\frac{2}{3}}} \right) \quad (52)$$

Next, we can transform (34) into (37) and then (38) by substituting

$$\begin{aligned} \gamma &= \bar{\eta} \\ b &= \frac{1}{10} \\ r_t &= \mathbb{E}(f(x^r) - f^*) \\ s_t &= \mathbb{E}(\|\nabla f(x^r)\|^2) \\ c_1 &= \frac{2L\sigma^2}{KL} + 2L \frac{M-K}{K(M-1)} \zeta^2, \\ c_2 &= \frac{15}{8} \frac{L^2\sigma^2}{KL} + \frac{15}{8} \frac{L^2\zeta^2}{K} \end{aligned}$$

and

$$\frac{1}{d} = \frac{1}{6L(1 + \frac{\beta^2}{K})}$$

Using this, we get

$$\begin{aligned} \min_{0 \leq r \leq R} \mathbb{E}[\|\nabla f(x^r)\|^2] &\leq \frac{10(f(x^0) - f^*)}{\bar{\eta}R} + \frac{20L\bar{\eta}\sigma^2}{KL} \\ &+ \frac{20L\bar{\eta}\zeta^2(M-K)}{K(M-1)} + \frac{75L^2\bar{\eta}^2\sigma^2}{4KL} + \frac{75L^2\bar{\eta}^2\zeta^2}{4K} \end{aligned} \quad (53)$$

When

$$\bar{\eta} = \min \left(\frac{1}{L(1 + \frac{\beta^2}{K})}, \frac{A^{\frac{1}{2}}}{c_1^{\frac{1}{2}} R^{\frac{1}{2}}}, \frac{A^{\frac{1}{3}}}{c_2^{\frac{1}{3}} R^{\frac{2}{3}}} \right) \quad (54)$$

and

$$A = f(x^0) - f^* \quad (55)$$

we can bound

$$\min_{0 \leq r \leq R} \mathbb{E}[\|\nabla f(x^r)\|^2] \quad (56)$$

by

$$\mathcal{O} \left(\frac{L\sigma^2 A^{\frac{1}{2}}}{\sqrt{KLR}} + \sqrt{1 - \frac{K}{M}} * \frac{(L\zeta^2 A)^{\frac{1}{2}}}{\sqrt{KR}} + \frac{(L^2\sigma^2 A^2)^{\frac{1}{3}}}{(KL)^{\frac{1}{3}} R^{\frac{2}{3}}} + \frac{(L^2\zeta^2 A^2)^{\frac{1}{3}}}{K^{\frac{1}{3}} R^{\frac{2}{3}}} + \frac{LA(1 + \frac{\beta^2}{K})}{R} \right) \quad (57)$$

Equation (44) is our convergence bound.

Theorem 1. *Let Assumptions (1), (6), (15), and (18) hold. Then the proposed update satisfies the convergence bound*

$$\mathcal{O} \left(\frac{L\sigma^2 A^{\frac{1}{2}}}{\sqrt{KLR}} + \sqrt{1 - \frac{K}{M}} * \frac{(L\zeta^2 A)^{\frac{1}{2}}}{\sqrt{KR}} + \frac{(L^2\sigma^2 A^2)^{\frac{1}{3}}}{(KL)^{\frac{1}{3}} R^{\frac{2}{3}}} + \frac{(L^2\zeta^2 A^2)^{\frac{1}{3}}}{K^{\frac{1}{3}} R^{\frac{2}{3}}} + \frac{LA(1 + \frac{\beta^2}{K})}{R} \right) \quad (58)$$

This convergence rate is dominated by $\mathcal{O}(\frac{1}{\sqrt{KLR}})$.

B. Note on notation

The notation in this appendix follows the original proof draft and is preserved here to keep the full derivation intact. The notation used in the main paper is lighter and presentation-oriented, whereas this appendix retains the detailed symbols needed for the full proof steps.

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